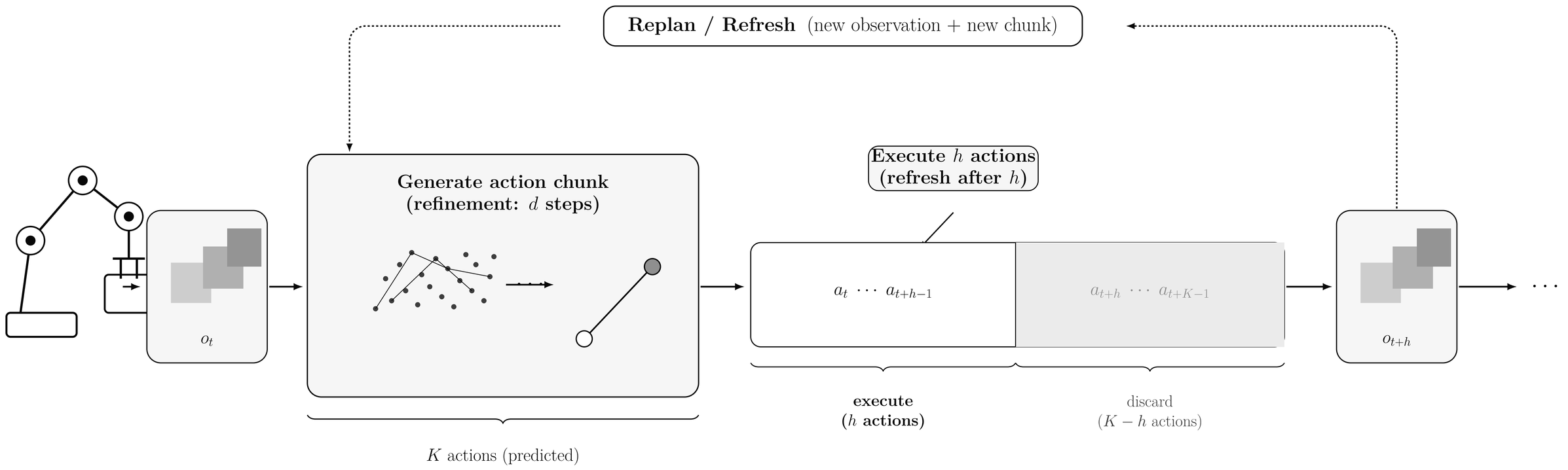
Refinement vs. Refresh

Compute Allocation for Closed-Loop Robot Policies

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Chunked Closed-Loop Policy



K: chunk length
(total predicted actions)

d: refinement
(denoising steps)

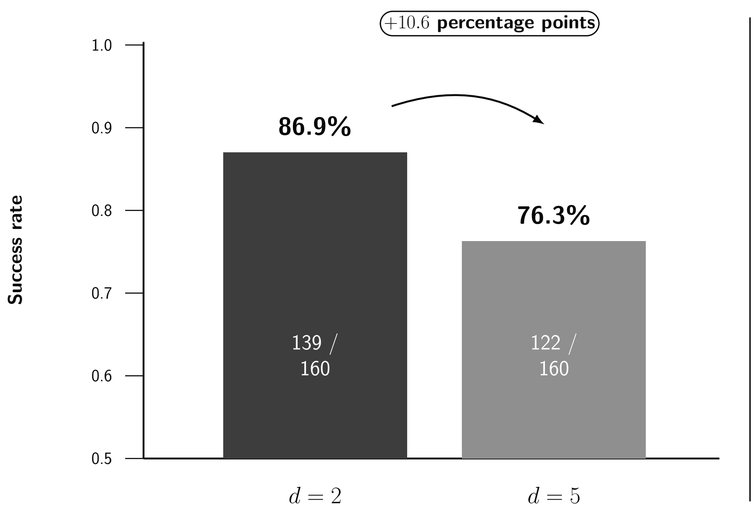
h: refresh
(actions before replanning)

Setup (held-out)

SmolVLA (frozen)	$K = 50,\ d \in \{2, 5\}$	
LIBERO (simulation)	$h \in \{2, 5\}$	8 tasks, 1,120 attempts

? Under a fixed inference budget, should we allocate compute to additional denoising steps (**refinement**) or to more frequent observation-conditioned replanning (**refresh**)?

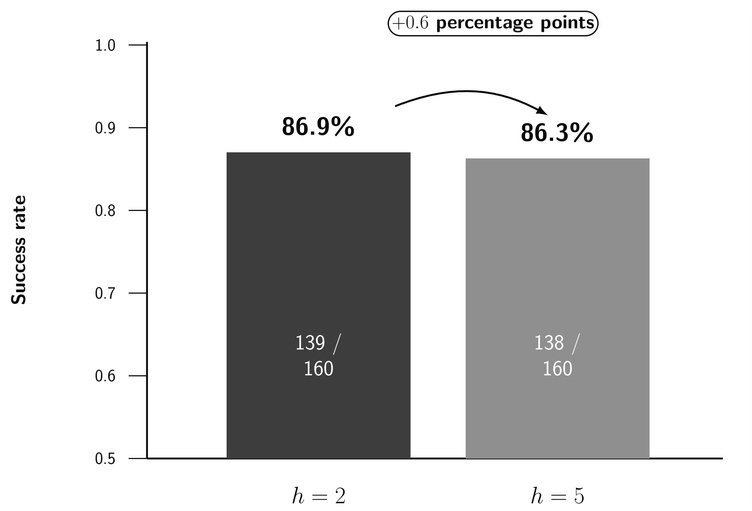
Result 1: Fewer Denoising Steps Improve Success



Takeaway

Reducing denoising from 5 to 2 steps improves success by **10.6 pp** while using **5.9% less** arithmetic per control step.

Result 2: Less Frequent Refresh Is Much Cheaper



Takeaway

At $d = 2$, executing 5 actions rather than 2 before replanning gives similar observed success (138/160 vs. 139/160) with **60.7% less** arithmetic per attempt.

Why This Matters

Inference compute is a control decision. For this checkpoint, fewer denoising steps improved success; executing more of each predicted chunk cut arithmetic sharply with similar observed success.

Interpretation & Limitations

Takeaways

- Fewer denoising steps improved success at $h = 2$.
- At $d = 2$, longer execution was far cheaper with similar observed success.

Limitations

- One SmolVLA checkpoint; synchronous simulation.
- Prefix comparison is descriptive; equivalence is not established.
- Arithmetic does not establish physical-robot latency.